1 Inner Products

We defined a vector space V be abstracting the properties of addition and scalar multiplication on $\mathbb{R}^n$. We have so far made no mention of the dot product, which plays a prominent role in geometry in $\mathbb{R}^n$. We now abstract the properties of the dot product to obtain an "inner product" on a vector space V, which will allow us to introduce geometric notions of length and angle in V.

The dot product is a mapping $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ which takes two vectors $\mathbf{x} = (x_1, \ldots, x_n)$ and $\mathbf{y} = (y_1, \ldots, y_n)$ to the real number

$$\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

It is straightforward to verify that the dot product on $\mathbb{R}^n$ has the following properties for all $\mathbf{x}, \mathbf{y}, \mathbf{y}_1, \mathbf{y}_2 \in \mathbb{R}^n$ and all $c \in \mathbb{R}$:

(i) $\mathbf{x} \cdot \mathbf{x} \geq 0$, where $\mathbf{x} \cdot \mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$;

(ii) $(c\mathbf{x}_1 + \mathbf{x}_2) \cdot \mathbf{y} = c\mathbf{x}_1 \cdot \mathbf{y} + \mathbf{x}_2 \cdot \mathbf{y}$;

(iii) $\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot \mathbf{x}$.

(iv) The the length $||\mathbf{x}||$ of $\mathbf{x} \in \mathbb{R}^n$ is given by $||\mathbf{x}|| = \sqrt{\mathbf{x} \cdot \mathbf{x}}$.

Exercise 1.1. Show that properties (ii) and (iii) imply that $\mathbf{x} \cdot (c\mathbf{y}_1 + \mathbf{y}_2) = c\mathbf{x} \cdot \mathbf{y}_1 + \mathbf{x} \cdot \mathbf{y}_2$.

Properties (ii) and (iii) above say that the dot product is a symmetric bilinear form on $\mathbb{R}^n$. A bilinear form satisfying property (i) is said to be *positive-definite*. We abstract these properties to define an inner product on a real vector space.

Definition 1.2 (Inner product on a real vector space)**.** Let V be a real vector space. We define an *inner product* on V to be a positive-definite symmetric bilinear form on V. We denote the inner product of two vectors $v, w \in V$ by (v, w).

Definition 1.2 does not quite work if V is a complex vector space if we want property (iv) to also hold. Recall that the magnitude of a complex number $z = x + iy \in \mathbb{C}^n$ can be written as

$$||z|| = \sqrt{z\overline{z}} = \sqrt{(x+iy)(x-iy)} = \sqrt{x^2+y^2}.$$

On $\mathbb{C}^n$, the definition of $||z||$ for $z = (z_1, \ldots, z_n)$ is

$$||z|| = \sqrt{\sum_{i=1}^{n} z_i \overline{z}_i}.$$

We therefore define a dot product on $\mathbb{C}^n$ by

$$z \cdot w = \sum_{i=1}^{n} z_i \overline{w}_i$$

so that $||z|| = \sqrt{z \cdot z}$. It is straightforward to check that this dot product on $\mathbb{C}^n$ has the same properties (i), (ii), and (iv) as the dot product on $\mathbb{R}^n$. However, it does not have property (iii). Instead, we see that

$$w \cdot z = \sum_{i=1}^{n} w_i \overline{z}_i = \sum_{i=1}^{n} \overline{z}_i w_i = \sum_{i=1}^{n} \overline{z_i \overline{w}_i} = \overline{\sum_{i=1}^{n} z_i \overline{w}_i} = \overline{z \cdot w}$$

Exercise 1.3. Show that the dot product on $\mathbb{C}^n$ defined above satisfies

$$w \cdot (cz_1 + z_2) = \overline{c} w \cdot z_1 + w \cdot z_2$$

for all $z_1, z_2 \in \mathbb{C}^n$ and all $c \in \mathbb{C}$.

The dot product on $\mathbb{C}^n$ is a mapping from $\mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ which is linear in the first entry but, by Exercise 1.3, is conjugate-linear in the second entry. This dot product is therefore said to be a *sesquilinear form* on $\mathbb{C}^n$ (since "sesqui" means "one and a half times").

Definition 1.4 (Inner product on a complex vector space)**.** Let V be a complex vector space. We define an *inner product* on V to be a positive-definite sesquilinear form on V.

Example 1.5. The dot products $(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} x_i y_i$ and $(\mathbf{z}, \mathbf{w}) = \sum_{i=1}^{n} z_i \overline{w}_i$ are of course inner products on $\mathbb{R}^n$ and $\mathbb{C}^n$. Since the $\overline{w} = w$ if w is real, we can just say that $(\mathbf{z}, \mathbf{w}) = \sum_{i=1}^{n} z_i \overline{w}_i$ is an inner product on F^n.

Definition 1.6 (Inner product space)**.** An *inner product space* is a vector space V together with an inner product on V.

Exercise 1.7. Show that if $c_1, \ldots, c_n$ are positive real numbers, then $(\mathbf{z}, \mathbf{w}) = \sum_{i=1}^{n} c_i z_i \overline{w}_i$ is an inner product on F^n.

Exercise 1.8. Let $V = \mathscr{C}([-1,1])$ be the vector space of continuous real-valued functions on the interval $[-1,1]$. Show the $(f,g) = \int_{-1}^{1} f(x)g(x)dx$ defines an inner product on V.

Exercise 1.9. Show that the following maps define inner products on $\mathscr{P}(\mathbb{R})$.

(a) $(p,q) = p(0)q(0) + \displaystyle\int_{-1}^{1} p'(x)q'(x)dx$

(b) $(p,q) = \displaystyle\int_{0}^{\infty} p(x)q(x)e^{-x}dx$

Definition 1.10 (Orthogonality)**.** Two vectors u, v in an inner product space V are said to be *orthogonal* if $(u, v) = 0$.

Exercise 1.11. Prove that $(v, 0) = (0, v) = 0$ for every $v \in V$.

Theorem 1.12. *If u, v are orthogonal vectors in an inner product space V, then u and v satisfy the Pythagorean theorem:*

$$||u+v||^2 = ||u||^2 + ||v||^2.$$

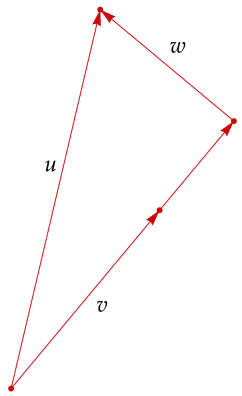
Exercise 1.13. Prove Theorem 1.12.

Proposition 1.14 (Orthogonal decomposition)**.** *Fix a vector v in an inner product space V. Then any $u \in V$ can be decomposed uniquely as*

$$u = u_{\parallel} + u_{\perp},$$

where $u_{\parallel}$ is a vector parallel to v and $u_{\perp}$ is perpendicular to v.

Exercise 1.15. Prove Proposition 1.14.



An orthogonal decomposition:
u expressed as a scalar multiple of v plus a vector orthogonal to v.

We now prove some important inequalities.

Proposition 1.16 (Cauchy-Schwarz inequality)**.** *For $u, v \in V$,*

$$|(u,v)| \leq ||u||\ ||v||.$$

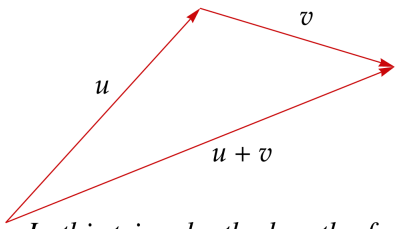
Exercise 1.17. Prove the Cauchy-Schwarz inequality. When does equality hold?

Exercise 1.18. The angle between two vectors (thought of as arrows with initial point at the origin) in $\mathbb{R}^2$ or $\mathbb{R}^3$ can be defined geometrically. The angle θ between two nonzero vectors $\mathbf{x}$, $\mathbf{y}$ in $\mathbb{R}^n$ is defined to be $\theta = \arccos\left(\frac{(\mathbf{x},\mathbf{y})}{||\mathbf{x}||\ ||\mathbf{y}||}\right)$. Explain why the Cauchy-Schwarz inequality is needed to show that this definition makes sense.

Exercise 1.19. Prove the *Triangle Inequality*: For $u, v \in V$,

$$||u+v|| \leq ||u|| + ||v||.$$

When does equality hold?

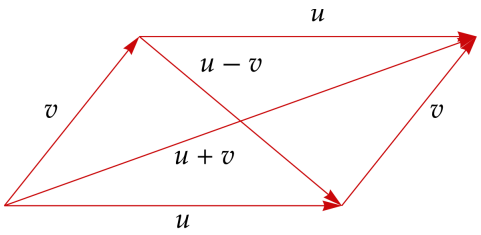


In this triangle, the length of $u + v$ is less than the length of u plus the length of v.

Exercise 1.20. Prove that if $u, v \in V$, then $\big|\,||u|| - ||v||\,\big| \leq ||u - v||$. This is called the *Reverse Triangle Inequality*.

Exercise 1.21. Prove the *Parallelogram Law*: For $u, v \in V$,

$$||u+v||^2 + ||u-v||^2 = 2(||u||^2 + ||v||^2).$$



The diagonals of this parallelogram are $u + v$ and $u - v$.

Exercise 1.22. If $u, v \in V$ and $||u|| = 3$, $||u+v|| = 4$, $||u-v|| = 6$, then what is $||v||$?

Exercise 1.23. Show that if $u, v \in V$, then $||u+v||\ ||u-v|| \leq ||u||^2 + ||v||^2$.